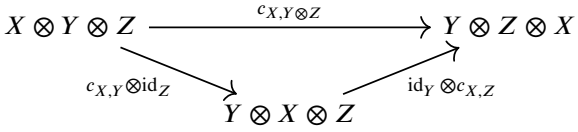
Representation Theory 2

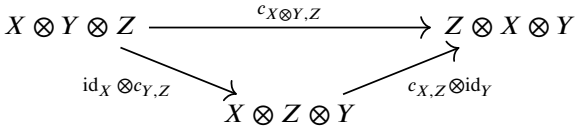
So Murata

WiSe 25/26, University of Bonn

0.1 Braidings

Definition 0.1. *A braiding/braided category is a pair $(\mathscr{C}, c)$ where $\mathscr{C}$ is a monoidal category and $c = (c_{X,Y} : X \otimes Y \to Y \otimes X)$ is a ntural isomorphism such that for any $X, Y, Z \in \mathscr{C}$, we have the commudatie diagrams.*





The braiding c is called symmetric if $c^2 = \mathrm{id}$. That is for any $X, Y \in \mathscr{C}$ we hae,

$$c_{X,Y} = c_{Y,X}^{-1}.$$

Lemma 0.1. *Let $(\mathscr{C}, c)$ be a braided category. We have the following statements.*

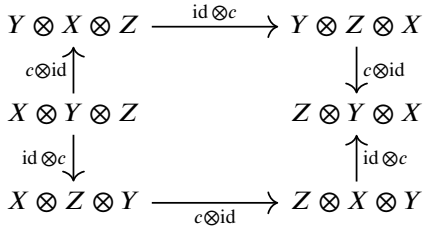
1). For $X \in \mathscr{C}$,

$$c_{X,1} = \lambda_X^{-1}\rho_X, c_{1,X} = \rho_X^{-1}\lambda_X = c_{X,1}^{-1}.$$

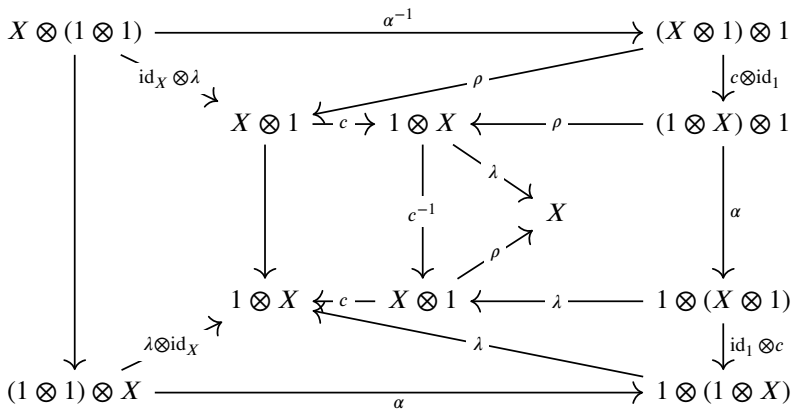
2). Young-Baxter equation holds. That is,

$$(c \otimes \mathrm{id})(\mathrm{id} \otimes c)(c \otimes \mathrm{id}) = (\mathrm{id} \otimes c)(c \otimes \mathrm{id})(\mathrm{id} \otimes c).$$

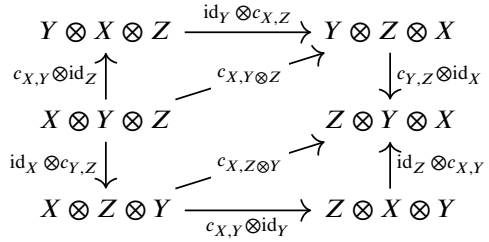
This is represented in the commutative diagram as follows.



Proof. In the view of Theorem **??**, and use c is a natural isomorphism, one has,



This proves the first assertion. For the second assertion,



□

Definition 0.2. *For $n \in \mathbb{Z}_{\geq 0}$, we define,*

Remark 0.1. *Let $\mathscr{C}$ be a $\mathbb{K}$-linear category and M be a set of objects. Then the full-subcategory of $\mathscr{C}$ of M is generally neither additive nor idempotent complete. However, let $\langle M \rangle_{\oplus}$ be the full subcategory of $\mathscr{C}$ on direct sums of objects from M. Then it is additive. Also $\langle M \rangle_{\overset{\subseteq}{\oplus}}$ the full subcategory of $\mathscr{C}$ that consists of on direct summands of direct sums of objects in M. It's additive and idempotent complete.*

Remark 0.2. *Let $\mathscr{C}$ be an additive, idempotent complete $\mathbb{K}$-linear category. Let M be a set of objects in $\mathscr{C}$. Generally the full subcategory of M is neither additive nor idempotent complete.*

Definition 0.3. *Let $\mathscr{C}$ be an additive, idempotent, $\mathbb{K}$-linear category and M be a set of objects in $\mathscr{C}$. We define,*

1). $\langle M \rangle_{\oplus}$ as the full subcategory of $\mathscr{C}$ with objects direct sums of objects from M.

2). $\langle M \rangle_{\overset{\subseteq}{\oplus,\oplus}}$ as the full subcategory of $\mathscr{C}$ with objects direct summands of direct sums of objects in M.

Proposition 0.1. *$\langle M \rangle_{\oplus}$ is additive and $\langle M \rangle_{\overset{\subseteq}{\oplus,\oplus}}$ is additive and idempotent complete.*

Proof. Additivity is clear in both cases. Let $X \in \langle M \rangle_{\overset{\subseteq}{\oplus},\overset{\subseteq}{\oplus}}$ and $e \in \mathrm{End}(X)$ be an idempotent. In $\mathscr{C}$, we have

$$\varphi : X \overset{\cong}{\to} X_1 \oplus X_2.$$

By definition, we can assume that $X_1, X_2 \in \langle M \rangle_{\overset{\subseteq}{\oplus},\overset{\subseteq}{\oplus}}$. Thus it is idempotent complete.

□

Lemma 0.2. *Let $\mathscr{C}$ be a $\mathbb{K}$-linear category, we have the following statements.*

1). $\mathscr{C}^{\oplus}, \mathscr{C}^{\overset{\subseteq}{\oplus},\oplus}$ are additive $\mathbb{K}$-linear.

2). $\mathscr{C}^{\overset{\subseteq}{\oplus}}, \mathscr{C}^{\overset{\subseteq}{\oplus},\overset{\subseteq}{\oplus}}$ are idempotent complete.

3). $\mathscr{C}$ is additive then $\mathscr{C} \cong \mathscr{C}^{\oplus}$ as $\mathbb{K}$-linear category.

4). $\mathscr{C}$ is idempotent complete then $\mathscr{C} \cong \mathscr{C}^{\overset{\subseteq}{\oplus}}$ as $\mathbb{K}$-linear categories.

Proof. For **1)**, the additiity is clear. For **2)**, let $(X, e) \in \mathscr{C}^{\overset{\subseteq}{\oplus}}$ and $efe \in \mathrm{End}((X, e))$ be an idempotent. Then $(X, efe) \in \mathscr{C}^{\overset{\subseteq}{\oplus}}$. Let

$$p = efe\,\mathrm{id}_X\, e \in \mathscr{C}^{\overset{\subseteq}{\oplus}}((X, e), (X, efe)), q = e\,\mathrm{id}_X\, efe \in \mathscr{C}^{\overset{\subseteq}{\oplus}}((X, efe), (X, e)).$$

Then we have,

$$pq = efeeefe = efe\,\mathrm{id}_X\, efe = \mathrm{id}_{(X,efe)}, qp = eefeefee = eefee = efe.$$

By Lemma **??**, this completes the proof.

3). is clear by setting $F : \mathscr{C} \to \mathscr{C}^{\oplus}$ be such that $F((X, p_i, q_i)) = (X_i)$. Remains to show that $\mathscr{C} \cong \mathscr{C}^{\overset{\subseteq}{\oplus}}$ for idempotent complete category $\mathscr{C}$. Let $F : \mathscr{C} \to \mathscr{C}^{\overset{\subseteq}{\oplus}}$ be such that $F(X) \to (X, \mathrm{id}_X)$. Clearly, this is fully-faithful. Remains to show this is essentially surjective. For $(X, e) \in \mathscr{C}^{\overset{\subseteq}{\oplus}}$, again by Lemma **??**, there is $Y \in \mathscr{C}$ such that

$$\exists p : X \to Y, q : Y \to X, qp = e, pq = \mathrm{id}_Y\,.$$

Thus we have $pe \in \mathscr{C}^{\overset{\subseteq}{\oplus}}((X, e), (Y, \mathrm{id}_Y))$ and $eq \in \mathscr{C}^{\overset{\subseteq}{\oplus}}((Y, \mathrm{id}_Y), (X, e))$. Then

$$peeq = peq = pqpq = \mathrm{id}_Y = \mathrm{id}_{(Y,\mathrm{id}_Y)},$$
$$eqpe = e = \mathrm{id}_{(X,e)}\,.$$

□